RESEARCH COMPILATION 2026

Beyond the LLM

2026

World models, active inference, post-transformer architectures, diffusion language models, agents, self-improvement and scalable oversight.

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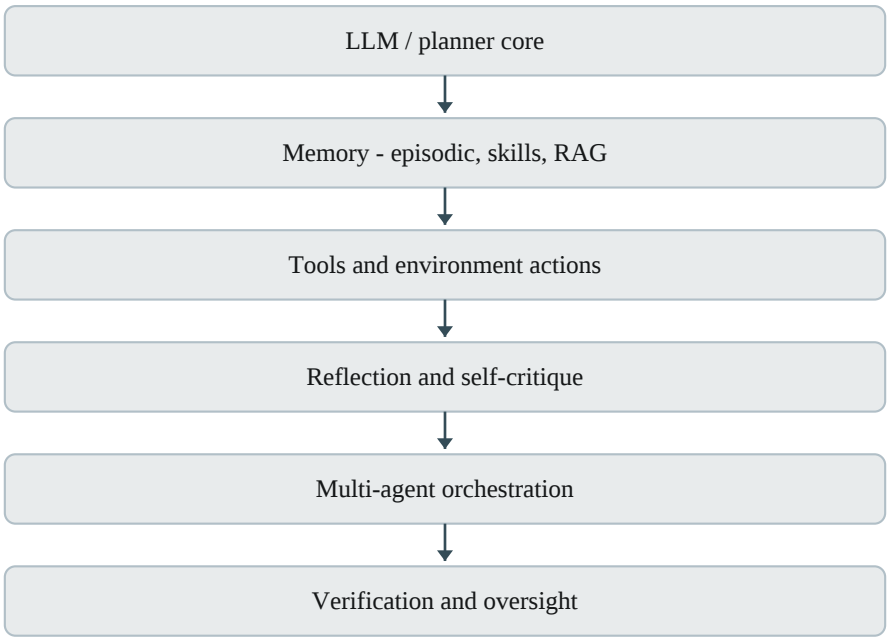
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Frontiers: New Types of Intelligence

Where the next intelligence comes from: the paradigm map beyond next-token prediction - planning in learned world models, self-play, and collective agent systems.

Beyond the LLM - the through-line



Part I - Distilled Findings

F1 - World models: plan in imagination, act in reality

Implication for building intelligence: LLMs predict tokens; agents need to predict CONSEQUENCES. World-model architectures are how you get consequence-prediction, and they are the substrate into which reasoning-style test-time search (domain 03, F5) naturally generalizes.

F2 - Active inference: one objective for perception, action, and curiosity

Implication for building intelligence: Reward design is the hardest part of RL systems (reward hacking, domain 03 F3). Active inference replaces hand-crafted curiosity bonuses with a term that falls out of the math - a principled alternative worth piloting wherever you currently write reward functions.

F3 - Attention is associative memory: the energy-based lineage returns

Implication for building intelligence: If you're designing "beyond transformer" systems, the attention-as-memory frame tells you WHAT attention is FOR (storage + retrieval), so you can replace it deliberately (e.g., with learned write/forget policies) rather than by kernel substitution.

F4 - The post-transformer zoo: each candidate encodes a different bias

Implication for building intelligence: Match architecture to constraint: long-stream/edge -> Mamba-family; symbolic/scientific discovery -> KAN; adaptive-at-inference -> TTT/Titans. Production trend (2025-26) is hybrids - attention layers for retrieval precision spliced into SSM layers for cheap context.

F5 - Diffusion language models break the serial-generation constraint

Implication for building intelligence: Generation speed becomes a hardware-parallel axis; bidirectional context and native infilling/editing come free. If autoregressive economics bind you (latency, cost), diffusion LMs are now a real alternative - the first crack in the AR orthodoxy since 2017.

F6 - Biological substrates: intelligence per watt, not per FLOP

Implication for building intelligence: The intelligence market is splitting: cloud-scale reasoning vs. edge/embodied sensing. For the second class, train small and deploy on event-driven hardware; the reservoir/liquid lesson (random structure + tiny trained readout) remains the cheapest usable pattern.

F7 - Neural-symbolic hybrids own systematic compositionality

Implication for building intelligence: For domains with verifiable structure (law, medicine, engineering, math), wrap LLM perception with explicit program/logic layers instead of hoping scale induces rigor. This is also the pragmatic architecture for ARC-class benchmarks.

F8 - Open-endedness: objectives deceive; novelty discovers

Implication for building intelligence: For self-improving agents (F11) the environment generator is as important as the learner: co-evolve tasks with solvers, keep a population (diversity), and don't over-anchor on a single metric - or the system will Goodhart it (domain 03, F3).

F9 - Collective intelligence: from emergent protocols to agent societies

Implication for building intelligence: The unit of intelligence is shifting from model to SYSTEM. Your leverage points are the protocol (what agents can say), the topology (who talks to whom - small-world, domain 04 F9), and the verification gate (who checks claims - F12 below).

F10 - The 2026 agent stack is a settled pattern

Implication for building intelligence: Don't invent agent architecture from scratch - instantiate this stack and spend your novelty budget on the memory format and the verifier quality, which is where systems actually differentiate. Every component descends from a specific paper; cite them when you build.

F11 - Self-improvement loops are real - and are the sharpest double-edged object in the field

Implication for building intelligence: This is the direction of travel for frontier labs. Two engineering duties follow: (1) verifier integrity is now the whole safety surface - who checks the checker; (2) watch for model collapse / distribution narrowing (domain 01, F8's data-quality lessons apply recursively).

F12 - Scalable oversight is a measurable engineering problem now

Implication for building intelligence: Any self-improving or agent system you deploy needs an oversight design: an adversarial check layer (debate-style), graded evaluators, and monitoring for the failure mode where the system learns what the checker rewards, not what you meant (Goodhart - domains 01 F13, 03 F3).

F13 - Generality is measurable - and still unsolved: the ARC line

Implication for building intelligence: Choose your target explicitly: skill (train bigger, distill) vs. generality (sample efficiency, program induction, priors). Benchmarks that separate them (ARC-class, procedurally generated) are the only trustworthy compass - LLM leaderboard wins increasingly measure skill, not intelligence.

Part II - The Canon

46 landmark papers in chronological order. Full entries with links and keywords live in the corpus folder 05_frontiers-new-intelligence/papers.md.

Year	Paper / Authors	Core idea	Why it matters
1997	**Networks of Spiking Neurons: The Third Generation of Neural Network Models** Wolfgang Maass	Neurons that communicate by discrete spikes are computationally more powerful than sigmoid units - even with the same analog noise - when time is used as a resource.	The theoretical charter for event-driven, sparse, energy-cheap computation; every neuromorphic hardware claim descends from it. (Cross-ref domain 04 for the neuroscience side; not yet covered there as of 2026-10.)
2001	**The "Echo State" Approach to Analysing and Training Recurrent Neural Networks** Herbert Jaeger	Keep a large random recurrent reservoir fixed and train only a linear readout - the "echo state" property makes the reservoir a fading memory of its input history.	Established reservoir computing: dynamics as a fixed feature map, learning confined to a cheap output layer - the low-energy predecessor of today's frozen-backbone + small-adapter designs.

Year	Paper / Authors	Core idea	Why it matters
2011	**Abandoning Objectives: Evolution Through the Search for Novelty Alone** Joel Lehman, Kenneth O. Stanley	Searching only for behavioral novelty - with no objective function at all - sometimes finds solutions that objective-driven search never reaches (e.g., the deceptive-maze robot).	Formalizes that objectives can actively mislead: stepping stones to intelligence may only be discoverable by open-ended, novelty-seeking search - a design principle for generating what we cannot specify.
2012	**Perceptions as Hypotheses: Saccades as Experiments** Karl Friston, Rick Adams, Laurent Perrinet, Michael Breakspear	Under active inference, perception and action both minimize the same free-energy functional - the brain literally acts to test its perceptual hypotheses.	The founding statement that one objective (expected free energy) can drive perception, planning, and curiosity - a template for agents that need no separate reward function. Kept short here; neuroscience context in domain 04.
2016	**Hybrid Computing Using a Neural Network with Dynamic External Memory (DNC)** Alex Graves, Greg Wayne, Ivo Danihelka, et al. (DeepMind)	A neural controller with a differentiable, addressable external memory matrix it learns to read, write, and allocate - inference and manipulation by pure gradient descent.	Memory as an explicit architectural component (not emergent state) - the ancestor of RAG indexes, agent episodic stores, and test-time-memory architectures.
2016	**Learning to Reinforcement Learn** Jane X. Wang, Zeb Kurth-Nelson, Dhruva Tirumala, et al. (DeepMind)	An RNN meta-trained across many tasks learns implicit RL algorithms in its recurrent dynamics - including exploration strategies adapted to each new task at test time.	Demonstrates that learning-to-learn can emerge in weights + activation state rather than hand-designed updates; the pre-LLM statement of "pretraining then adapting" for decision-making.
2016	**HyperNetworks** David Ha, Andrew Dai, Quoc V. Le	One network generates the weights of another, so model structure itself becomes a learned, dynamically adaptable function of context.	Weights stop being fixed parameters and become *state* - the conceptual root of fast weights, hypernetwork adapters, and test-time weight adaptation.
2017	**Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks (MAML)** Chelsea Finn, Pieter Abbeel, Sergey Levine	Optimize initial parameters explicitly so that a few gradient steps on a new task yield good performance - learn an initialization, not a solution.	The canonical formalization of "learn to adapt in k steps"; the reference point every sample-efficient-adaptation scheme (including LLM-tuning-as-meta-learning analogies) is measured against.

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2017	**Evolution Strategies as a Scalable Alternative to Reinforcement Learning** Tim Salimans, Jonathan Ho, Xi Chen, et al. (OpenAI)	Black-box evolution of policy parameters, parallelized across thousands of gradient-free workers, matches deep RL on MuJoCo while sidestepping backprop-through-time entirely.	Learning without gradients is embarrassingly parallel and non-differentiable-stack-friendly - the fallback optimizer for world models, agents, and hardware where gradients do not flow.
2018	**World Models** David Ha, Jürgen Schmidhuber	Train a tiny VAE world model and an RNN predictor of its latent dynamics, then evolve a small controller inside hallucinated dreams instead of the real environment.	First striking demonstration that a *learned internal simulator* can be the training ground for behavior - the conceptual seed of Dreamer and of all latent-imagination agents.
2018	**Neural Ordinary Differential Equations** Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, David Duvenaud	Define network depth as a continuous ODE solved by standard numerical integrators, with gradients via the adjoint method at constant memory cost.	Makes *continuous time* a first-class modeling axis - the mathematical basis for liquid networks and for architectures that trade discrete layers for physical-time dynamics.

Year	Paper / Authors	Core idea	Why it matters
2018	**Loihi: A Neuromorphic Manycore Processor with On-Chip Learning** Mike Davies et al. (Intel)	A 128-core chip with asynchronous spiking neurons and programmable on-chip plasticity rules, consuming milliwatts on workloads that cost GPUs watts.	Physical proof that learning can run at 3-5 orders of magnitude lower energy than GPU training/inference - the hardware existence theorem for always-on embedded intelligence.
2018	**DeepProbLog: Neural Probabilistic Logic Programming** Robin Manhaeve, Sebastijan Dumančić, Angelika Kimmig, Thomas Demeester, Luc De Raedt	Embed neural networks as probabilistic predicates inside a differentiable logic program, so gradients flow through logical inference itself.	The cleanest mechanism for systematic compositionality: symbols guarantee rule structure while sub-symbolic nets supply perception - exactly the generalization pattern pure LLMs still miss.
2018	**Emergence of Linguistic Communication from Referential Games** Angeliki Lazaridou, Alexander Peysakhovich, Marco Baroni (with 2016 antecedents)	Two agents invent a discrete language to solve referential games; grounded, compositional-ish protocols emerge from pure cooperation pressure, but drift toward non-human languages without inductive biases.	The experimental base camp for emergent communication: tells you exactly which pressures (grounding, population, ease) make shared symbolic systems arise - and which make them alien.
2018	**AI Safety via Debate** Geoffrey Irving, Paul Christiano, Dario Amodei (OpenAI)	Two adversarial AI debaters with a human judging only the transcript can, in principle, supervise capabilities beyond the judge's own - amplifying oversight instead of capability alone.	The core mechanism candidate for supervising smarter-than-human systems: supervision scales with computation spent on debate, not with judge expertise.
2019	**Learning Latent Dynamics for Planning from Pixels (PlaNet) / [2020] Dream to Control (Dreamer)** Danijar Hafner, Timothy Lillicrap, et al.	PlaNet learns a recurrent state-space model from pixels and plans in it; Dreamer makes it sample-efficient by backpropagating value through imagined trajectories (latent imagination) with a replay of only real interactions.	Converts RL from trial-and-error in reality to *learning inside a learned model* - the efficiency route to agents that cannot afford millions of real mistakes.

Year	Paper / Authors	Core idea	Why it matters
2019	**The Neuro-Symbolic Concept Learner: Interpreting Scenes, Words, and Sentences from Natural Supervision** Jiayuan Mao, Chuang Gan, Pushmeet Kohli, Joshua Tenenbaum, Jiajun Wu	A visual parser turns pixels into object programs, a semantic parser turns questions into symbolic programs, and answering is program execution - no program annotations.	Demonstrates human-like data efficiency and zero-shot recombination on VQA by splitting perception (neural) from reasoning (symbolic) - the architecture pattern behind 2024-25 neuro-symbolic ARC solvers.
2019	**Paired Open-Ended Trailblazer (POET)** Rui Wang, Joel Lehman, et al.	Endlessly generate new environments and their solvers together, transferring agents between paired challenges to keep difficulty at each learner's frontier.	The first working engine of open-ended co-evolution of tasks and solutions - an automatic curriculum generator, and a concrete answer to "where does an endless training signal come from?".
2020	**Scaling Active Inference** Alex Tschantz, Beren Millidge, Anil K. Seth, Christopher L. Buckley	Deep active-inference agents (expected free energy = extrinsic value + epistemic value) match or beat model-free RL on standard domains and solve tasks RL cannot without reward shaping.	Shows active inference is an engineering-viable alternative to reward-based RL - intrinsic curiosity falls out of the objective rather than being bolted on.

Year	Paper / Authors	Core idea	Why it matters
2020	**Hopfield Networks is All You Need** Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, et al. (Hochreiter group)	Transformer attention is mathematically equivalent to one update step of a modern (exponential-capacity) Hopfield network retrieving a stored pattern.	Reframes attention as *associative memory retrieval* - connecting transformers to the Hopfield energy-based lineage (domain 01) and licensing energy-based, memory-centric alternatives to softmax attention.
2021	**Liquid Time-Constant Networks** Ramin Hasani, Mathias Lechner, et al. (MIT)	Neural ODEs whose time constants are states, so each neuron adapts its own clock to the input - closed-form variants (CfC) make them cheap enough for deployment.	Networks that natively handle irregular sampling and time-warped streams (robotics, wearables, edge sensing) - where fixed-clock discrete stacks waste computation.
2022	**A Path Towards Autonomous Machine Intelligence (JEPA position paper)** Yann LeCun	An architecture blueprint (Joint Embedding Predictive Architecture) in which agents predict in *abstract representation space*, not pixels, using a hierarchy of world models plus a configurator and intrinsic cost.	The most detailed published alternative to the generative-next-token path: prediction without generation, planning in latent space, energy-based inference - and it now anchors a real research program (I-JEPA, V-JEPA).
2022	**STaR: Bootstrapping Reasoning With Reasoning** Eric Zelikman, Yuhuai Wu, Jesse Mu, Noah D. Goodman (Stanford)	Generate rationales, keep only those that produce correct final answers, fine-tune on them, and re-generate - reasoning ability bootstraps itself upward from a few seeds.	The origin loop of LLM self-improvement: the STaR pattern (generate, filter by outcome, train, repeat) is the direct ancestor of rejection-sampling fine-tuning and RLVR-era reasoning training.
2022	**Measuring Progress on Scalable Oversight for Large Language Models** Samuel R. Bowman et al.	A unified evaluation suite (weak-strong pairings, sandwiching, debate, market-making) for the concrete question: can weak overseers extract truthful behavior from stronger models?	Turns oversight from philosophy into benchmark engineering - defines the measurable race between model capability and our ability to check it.

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2023	**Mastering Diverse Domains through World Models (DreamerV3)** Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, Timothy Lillicrap	A single fixed hyperparameter set masters 150+ tasks across benchmarks, and becomes the first agent to collect a Minecraft diamond from scratch, with no human data.	First evidence that one world-model-based learner can scale across wildly different domains - a concrete counter-proposal to "one benchmark per specialized RL stack".
2023	**Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture (I-JEPA)** Mahmoud Assran, Quentin Duval, Ishan Misra, et al. (Meta)	Predict representations of masked image regions from other regions, in embedding space - no pixel reconstruction, no hand-crafted augmentations.	First large-scale working instance of LeCun's blueprint: representation learning by latent prediction beats generative and contrastive pipelines, validating "predict abstractions, not pixels".
2023	**RWKV: Reinventing RNNs for the Transformer Era** Bo Peng, Eric Alcaide, et al.	An RNN that can be trained in parallel like a transformer (linear attention reformulated as a recurrence) and runs in constant memory at inference.	Proof that recurrence and parallel training are not mutually exclusive - a streaming, deployable alternative to quadratic attention born in the open-source community.
2023	**Hyena Hierarchy: Towards Larger Convolutional Language Models** Michael Poli, Stefano Massaroli, Eric Nguyen, et al.	Replace attention with long implicit convolutions gated by a small MLP, yielding subquadratic scaling to hundreds of thousands of context tokens.	Showed the attention kernel itself is replaceable at language-model scale - the long-convolution branch of the post-transformer zoo that Mamba later overshadowed.

Year	Paper / Authors	Core idea	Why it matters
2023	**Mamba: Linear-Time Sequence Modeling with Selective State Spaces** Albert Gu, Tri Dao	Make state-space parameters input-dependent (selective), so the model chooses what to remember or forget - transformer-class quality at linear time and constant per-token memory.	The flagship non-transformer backbone: streaming/edge inference, million-token contexts, and DNA/audio modeling - and the template the whole 2024+ SSM ecosystem (Jamba, hybrid stacks) builds on.
2023	**SpikeGPT: Generative Pre-trained Language Model with Spiking Neural Networks** Rui-Jie Zhu, Qihang Zhao, Guoqi Li, Jason K. Eshraghian	A GPT-style language model (up to 345M params) built on spiking RWKV neurons, generating text with binary event-driven activations at roughly 20x less energy than an equivalent dense model.	The bridge paper: modern generative LLM architecture rendered in spiking form - connects the LLM world to neuromorphic hardware deployment rather than toy tasks.
2023	**Generative Agents: Interactive Simulacra of Human Behavior** Joon Sung Park, Joseph O'Brien, Carrie Cai, et al. (Stanford)	25 LLM agents with memory streams, reflection, and planning in a sandbox town produce believable social behavior (party planning, relationship formation) evaluated by human raters.	The founding demo of society-scale simulation: LLMs as substrate for social-science experiments, agent economies, and synthetic populations - collective intelligence built from generative individuals.
2023	**Toolformer: Language Models Can Teach Themselves to Use Tools** Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, et al. (Meta)	A model self-annotates where inserting an API call (calculator, search, QA) would reduce its own perplexity, then fine-tunes on those insertions - tool use learned, not prompted.	The model becomes the dispatcher that decides *when* computation should leave its weights - the learning-side foundation of function calling and of agents that offload exact work to tools.

Year	Paper / Authors	Core idea	Why it matters
2023	**Reflexion: Language Agents with Verbal Reinforcement Learning** Noah Shinn, Federico Cassano, Beck Labash, et al.	Agents that fail a task store a *verbal* self-diagnosis in episodic memory and retry with that lesson in context - episodic control implemented in language instead of weights.	Cheap, gradient-free self-improvement: converting execution feedback into natural-language policy updates is now the standard inner loop of serious agent frameworks.
2023	**Voyager: An Open-Ended Embodied Agent with Large Language Models** Guanzhi Wang, Yuqi Xie, Yunfan Jiang, et al. (NVIDIA, Caltech)	An LLM agent in Minecraft writes its own actions as code, verifies them, and stores verified skills in a growing skill library that compounds across tasks without any gradient updates.	The skill-library pattern: continual learning as *accumulated executable artifacts* - the most copied mechanism for making agents improve monotonically in open worlds.
2023	**Weak-to-Strong Generalization: Eliciting Strong Capabilities With Weak Supervision** Collin Burns, Pavel Izmailov, Jan Hendrik Kirchner, et al. (OpenAI)	Fine-tuning a strong model on labels from a much weaker one recovers most of the strong model's capability - and sometimes beats the weak supervisor's own ceiling.	Empirical hope that humans can keep steering superhuman models via weak signals (if generalization conditions hold) - the working hypothesis behind 2025-26 oversight plans for self-improving systems.
2024	**Revisiting Feature Prediction for Learning Visual Representations from Video (V-JEPA)** Adrien Bardes, Quentin Garrido, Jean Ponce, et al. (Meta)	Extends JEPA to video: predict latent features of temporally masked frames; the model develops intuitive physics (object permanence, action prediction) without labels.	Moves world models toward the pretraining-scale regime - a video-prediction substrate that downstream agents and robotics stacks can build on.
2024	**xLSTM: Extended Long Short-Term Memory** Maximilian Beck, Korbinian Pöppel, Markus Spanring, et al. (Hochreiter group)	Modernizes the LSTM with exponential gating and matrix memory (sLSTM, mLSTM), restoring the classic cell's competitiveness with transformers and Mamba.	The recurrence line did not die in 2017 - a direct counter-demonstration that gated recurrence, properly upgraded, belongs in the architecture search space.

Year	Paper / Authors	Core idea	Why it matters
2024	**KAN: Kolmogorov-Arnold Networks** Ziming Liu, Yixuan Wang, Sachin Vaidya, et al.	Put learnable activation functions on edges (B-splines) instead of fixed activations on nodes, exploiting the Kolmogorov-Arnold representation theorem.	Small KANs discover symbolic, interpretable formulas where MLPs fit opaque surfaces - an architecture whose inductive bias points toward compositional and scientific discovery, not just benchmark loss.
2024	**Learning to (Learn at Test Time): RNNs with Expressive Hidden States (TTT layers)** Yu Sun, Xinhao Li, Karan Dalal, et al. (Stanford)	Make the RNN hidden state itself a small neural network whose update rule is a step of self-supervised learning - the sequence layer literally trains its memory during the forward pass.	Unifies fast weights, meta-learning, and long-context modeling: adaptation stops being a separate fine-tuning phase and becomes part of inference.
2024	**Multi-agent LLM ecosystems: swarms, MCP, agent-to-agent protocols** ecosystem development (no single paper)	A practical stack crystallizes around standard tool protocols (MCP, 2024), agent swarms/orchestration frameworks, and emerging agent-to-agent communication formats across vendors.	Multi-agent intelligence is becoming *infrastructure* rather than research demo - standard interfaces let heterogeneous agents compose, which is the precondition for agent economies and collective systems that outgrow any single model.

Year	Paper / Authors	Core idea	Why it matters
2024	**OpenAI o3 on ARC-AGI: reasoning models crack (part of) the generality benchmark** OpenAI (system milestone)	o3, trained with RL on verifiable tasks plus large test-time search, scored 75.7% (87.5% with 172x compute) on ARC-AGI-Pub - the first big jump on the benchmark that had resisted a decade of approaches.	Evidence that Chollet's skill-acquisition framing (domain 01) responds to RL + test-time search - while costing enormous compute per task, showing "generality" and "efficiency" are still far from joint. System-scale evidence, flagged as such.
2025	**Titans: Learning to Memorize at Test Time** Ali Behrouz et al. (Google)	A neural long-term memory module that *surprises-based* writes and forgets at test time, coordinated with short-term attention in a hybrid architecture that scales beyond 2M context.	Memory management (what to store, when to forget) becomes an architectural, learned-in-place mechanism - a direct answer to the context-window economics of long-horizon agents.
2025	**Large Language Diffusion Models (LLaDA)** Shen Nie, Fengqi Zhu, Chao Du, et al. (Ant Group, Renmin University)	An 8B-parameter diffusion LM trained from scratch with masked-diffusion pretraining and instruction tuning that rivals LLaMA3-8B on many benchmarks.	First convincing evidence that the *generative paradigm itself* (any-order parallel denoising vs left-to-right factorization) is swappable at language-model scale, with built-in infilling and controllability.
2025	**Commercial diffusion language models: Mercury and Gemini Diffusion** Inception Labs; Google DeepMind	Mercury (2025-02) ships the first commercial-scale diffusion LLM at 1000+ tokens/sec on H100s; Gemini Diffusion (2025-05) is Google DeepMind's research-preview diffusion model with comparable speed.	Parallel diffusion decoding turns generation speed from a serial-token constraint into a hardware-parallel one - the first deployed crack in autoregressive orthodoxy.
2025	**Absolute Zero: Reinforced Self-play Reasoning with Zero Data** Andrew Zhao, Yiran Wu, Yang Yue, et al. (Tsinghua LeapLab)	One model both *proposes* tasks (to maximize its own learnability) and solves them, verified by a code executor - AlphaZero-style self-play for reasoning with zero external data.	A working escape from the human-data bottleneck: the training signal is generated and checked by machine, pointing at open-ended self-improvement loops (and their unsupervised-error risks).

Year	Paper / Authors	Core idea	Why it matters
2025	**ARC-AGI-2: the persistent holdout** ARC Prize Foundation (benchmark milestone)	A redesigned ARC suite targeting the efficiency gap - tasks tuned so humans still solve most while frontier reasoning models score in the single to low double digits at release.	Even o3-class systems only partially crack ARC-AGI-1 at extreme cost, and ARC-AGI-2 re-opens the gap: a live yardstick separating "skill at scale" from general, *sample-efficient* intelligence (Chollet 2019, domain 01).

Full sources with links live in the corpus: papers.md / papers.json per domain.

Where to go next

The living version of this canon lives in the corpus folder for this domain: papers.md carries every entry with full links and keywords, findings.md holds the complete distilled findings, and the diagrams folder contains Mermaid sources that render in Obsidian and GitHub. To see how this domain connects to the rest of the atlas, read the cross-domain map in 00_master-atlas; to extend the corpus, follow the conventions in AGENTS.md and re-run the index and PDF build scripts in tools/.